Multi-target IC v3: k-term table (ic5), 22-32 bit groups

Multi-target IC v3 results: the k-term table (ic5) improves the multi-target IC in every cell; against rho it is
not established

Protocol: `PROTOCOL.md` (frozen in f71b42cc4f; amendment A1 after the rho-restart fix 75dbc70b36, before any holdout
result). ic5 parameters frozen from the tune set in `frozen_ic5.json` (mu = 1.0, k = 4) before the holdout.
Holdout: 44 fresh prime-order curves (indices 1..11 at 22, 24, 28, 32 bits), seeds 21 and 22, T in {1024, 4096,
16384}: **4488 runs, 0 errors, every target of every run verified**. Data: `results/holdout/rows.jsonl` (artifact
store), `results/holdout/runs.json` (compact counters in Git), `analysis.txt`, `summary.json`, `noverify.txt`.
Every number here is [pcc:unverified] (claim PCC-20261111-751c50) until the claim is signed. Independent validation:
`validation/VALIDATOR_REPORT.md` (verdict valid_with_amendments; every row replays exactly, every number recomputes;
the vs-rho headline is insufficient evidence, see below). The amendments it required are applied in this file.

Cost per target in GAE = [(c_I I + M + S)/(c_I + 3) + h (H + L)] / T. Each ratio is the median per-configuration
ratio (22 configurations per cell); ranges are over c_I in {3, 10, 30} and h in {0, 0.1, 1}. rhobest = per
configuration and price, the cheapest of the 12 pre-registered runs of this harness's anchored, fixed-database rho
family (rho, rho2, rho2x, rhoseq at kappa 0.5, 0.7, 1.0; no retention of finished walks, no distinguished points),
after the restart fix; bsgs2 = best of three mf.

Result

- **Validated: ic5 is cheaper than ic4, the multi-target IC baseline, in all 36 (cell, c_I) decisions**, by
  0.47x to 0.99x (largest at 32 b, T = 1024). Class: engineering improvement of the IC (AGENTS.md section 3) [pcc:unverified].
- **Distance remaining to BSGS:** ic5 is dearer than bsgs2 in all 36, 1.11x to 1.49x.
- **Against rho: not established.** ic5 is 0.85x to 0.98x the harness rho family (cheaper in all 36 under the
  frozen rule), but that reference lacks the standard multi-target rule of keeping the walk points of finished
  targets (Kuhn-Struik retention). The validator added retention to rho2x (`rho2xks`, one counted hash insert per
  retained point, 1056 runs, every target verified; `validation/ks_runs.json`): against it ic5 is **1.07x-1.34x
  at h = 0, 1.05x-1.18x at h = 0.1, 0.91x-0.98x at h = 1**, so the section 5a decision is unranked in all 36. A
  distinguished-point rho is estimated (model, not measured) to be cheaper at h = 1 as well. Retention gives ic5
  itself only 0-2%. No crossover against rho is claimed anywhere until a retention or distinguished-point rho is
  measured under a frozen rule.
- Even within the harness family, the fixed rho still pays avoidable overheads (restarts 2.2-3.0%, repeated
  lookups 1.5-3.5%, phase-1 merge waste; `validation/rho_instrumentation.json`); removing the first two leaves 30
  cheaper and 6 unranked (all six at c_I = 30).

Main table (as measured, against the pre-registered references)

bits	T	ic5/rhobest	ic5/bsgs2	ic5/ic4	ic4/rhobest	ic4 vs rhobest (c_I 3/10/30)
22	1024	0.919-0.963	1.275-1.361	0.830-0.894	1.071-1.130	dearer / dearer / dearer
22	4096	0.931-0.972	1.181-1.278	0.927-0.970	0.984-1.016	unranked / unranked / unranked
22	16384	0.931-0.982	1.112-1.197	0.969-0.992	0.960-0.990	cheaper / cheaper / cheaper
24	1024	0.915-0.975	1.344-1.405	0.786-0.840	1.125-1.213	dearer / dearer / dearer
24	4096	0.912-0.962	1.243-1.334	0.902-0.944	0.987-1.042	unranked / dearer / dearer
24	16384	0.925-0.968	1.158-1.255	0.958-0.984	0.954-0.988	cheaper / cheaper / cheaper
28	1024	0.875-0.959	1.430-1.447	0.634-0.662	1.360-1.502	dearer / dearer / dearer
28	4096	0.870-0.946	1.367-1.410	0.831-0.863	1.037-1.134	dearer / dearer / dearer
28	16384	0.880-0.945	1.283-1.358	0.922-0.949	0.944-1.019	cheaper / unranked / unranked
32	1024	0.874-0.975	1.469-1.491	0.473-0.483	1.826-2.060	dearer / dearer / dearer
32	4096	0.854-0.952	1.423-1.432	0.707-0.723	1.190-1.338	dearer / dearer / dearer
32	16384	0.848-0.942	1.374-1.402	0.862-0.879	0.974-1.090	unranked / dearer / dearer

Section 5a decisions (bootstrap 95% interval of the median ratio at all three h):

- ic5 vs ic4: cheaper in all 36. ic5 vs bsgs2: dearer in all 36.
- ic5 vs the harness rho family (rhobest): cheaper in all 36; not robust (above).
- ic4 vs rhobest: cheaper only at T = 16384 for 22 and 24 b (and 28 b at c_I = 3); dearer at every T = 1024 cell
  and from 28 b at T = 4096; at 32 b, T = 16384 unranked or dearer.
- ic4 vs bsgs2: dearer in all 36.

Without the verification phase that every arm shares (about 30 to 50 GAE per target), at I = 10 M
(`noverify.txt`): ic5/rhobest 0.883-0.962, ic5/bsgs2 1.42-1.53, ic4/rhobest 0.92-2.04.

Why ic5 is cheaper than ic4 (interpretation, from the counted phase split in `rows.jsonl`)

ic4 needs about 2m relations for m ~ sqrt(table) unknown factor-base logs (32 b, T = 1024: m = 1218, 2436
relations). ic5 with k = 4 reaches the same table size from m ~ table^(1/4) logs (48 to 67 at 32 b) and needs 48 to
111 relations per run (median 53.5), so its IC-specific phases (factor base, relation search, linear algebra) are
0.08% to 2.5% of its cost. What is left is the table (one shared inversion per two x-classes) and the paired target
trials, whose measured counts match n/(2K) (32 b, T = 1024: 1317-1495 trials per target across runs against
n/(2K) for each run's table). ic5's advantage over ic4 grows with the group and shrinks with T.

ic5's lead over the harness rho is therefore generic: a k-sum table with known logs would cost the same, and the
comparison with rho is between x-only shared-inversion tables and trials on one side and a walk family without
retention on the other. It does not change the exponent, and BSGS, which tiles its windows, stays 1.1x to 1.5x
cheaper (`../mt_bsgs_gap_20261111/README.md`).

Pre-registered predictions vs outcome

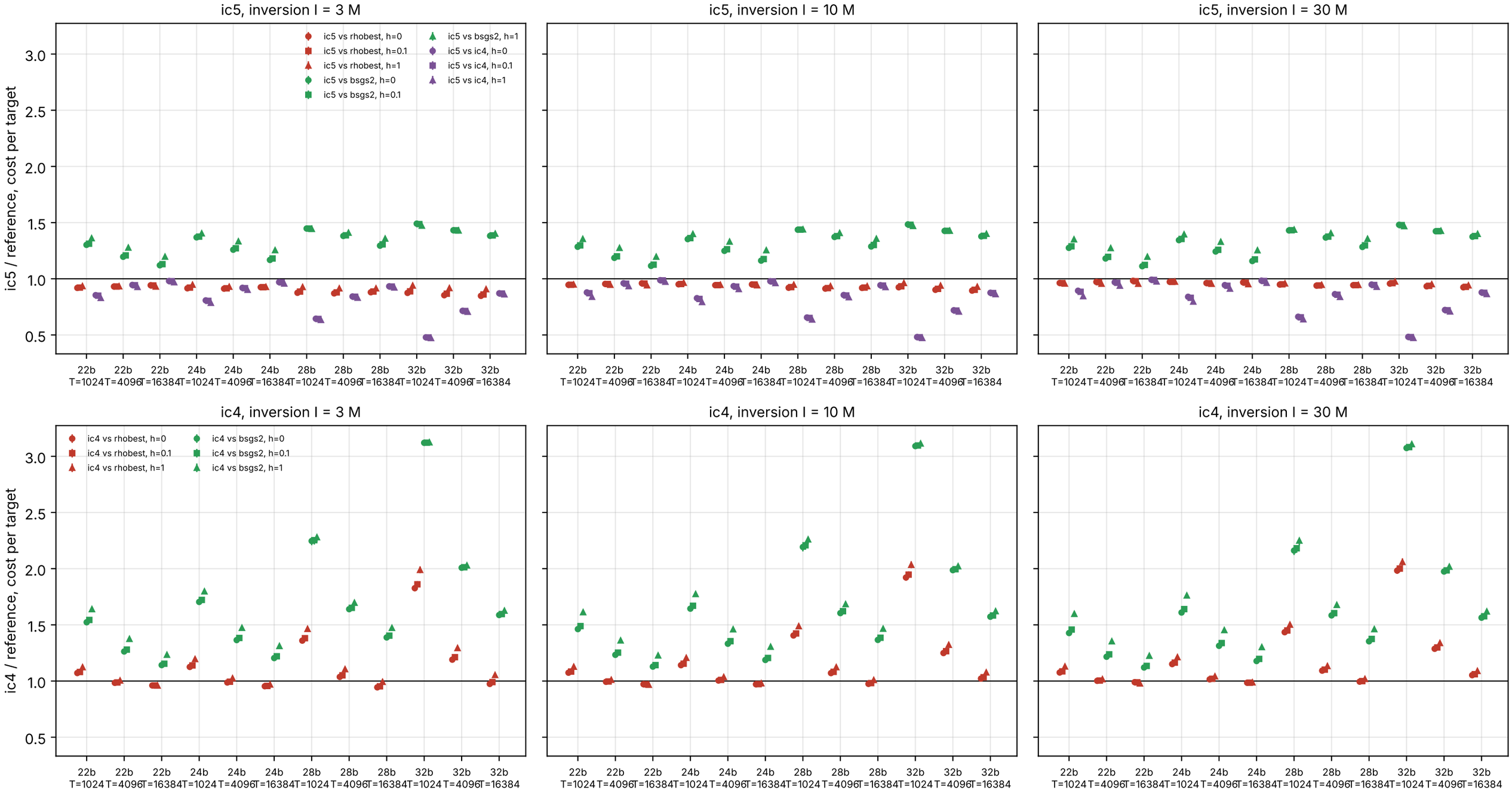
prediction	outcome
ic5/ic4 < 1 everywhere, largest gain at T = 1024 (about 0.85)	holds; the gain is larger than predicted and grows with size (0.47 at 32 b)
ic5/rhobest cheaper at T = 4096 and 16384 at every size	holds against the harness family; not robust to a retention rho
ic5/rhobest at T = 1024 about 0.92, cheaper or unranked	cheaper against the harness family (0.87-0.98); not robust
ic4/rhobest at 28 and 32 b: cheaper at 16384, dearer at 1024	dearer at 1024 holds; at 16384 only 28 b at c_I = 3 is cheaper, 32 b is unranked or dearer (wrong)
both dearer than bsgs2 everywhere	holds

Answers to the questions

1. (option 2, small T) ic5 lowers the multi-target IC's cost most at small T (0.47-0.89 of ic4 at T = 1024). Whether
   it is cheaper than rho at any T is not established: it is cheaper than this harness's rho family, but a rho with retention is
   cheaper at h = 0 and 0.1.
2. (option 1, larger groups) ic4's v2 pattern does not hold at 28 and 32 b: its relation phase grows like the square
   root of the table, and it falls behind even the harness rho except at T = 16384. ic5 keeps its margin over ic4
   and the harness rho with size.

Limits

Groups of 2^22 to 2^33; costs counted in field operations, not timed; memory unbounded for every arm; one local host (this container has no AWS identity, so not isolab-eks); the rho family is this harness's, not ecbench's, was fixed during this study (`../mt_rho_start_fix_20261111/`) and lacks retention and distinguished points. The comparison is per target in a batch of T targets on one curve; it says nothing about single-target ECDLP, and ECC2K-130 / the m = 83 binary curves are not engaged. The ic5 tune set chose k = 4, the largest k allowed; k = 5 was not tried. `rows.jsonl` files are moved to the artifact store by the store bot when the PR opens (`results/holdout/runs.json` keeps the counters in Git).


ic5, inversion I = 3 M
ic5, inversion I = 10 M
ic5, inversion I = 30 M
ic5 / reference, cost per target
ic5 vs rhobest, h=0
ic5 vs rhobest, h=0.1
ic5 vs rhobest, h=1
ic5 vs bsgs2, h=0
ic5 vs bsgs2, h=0.1
ic5 vs bsgs2, h=1
ic5 vs ic4, h=0
ic5 vs ic4, h=0.1
ic5 vs ic4, h=1
ic4, inversion I = 3 M
ic4, inversion I = 10 M
ic4, inversion I = 30 M
ic4 / reference, cost per target
ic4 vs rhobest, h=0
ic4 vs rhobest, h=0.1
ic4 vs rhobest, h=1
ic4 vs bsgs2, h=0
ic4 vs bsgs2, h=0.1
ic4 vs bsgs2, h=1
0.5
1.0
1.5
2.0
2.5
3.0
22b T=1024
22b T=4096
22b T=16384
24b T=1024
24b T=4096
24b T=16384
28b T=1024
28b T=4096
28b T=16384
32b T=1024
32b T=4096
32b T=16384